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Applying Return on Time Invested to Identify Obstacles That Yield Diminishing Returns

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Abstract

Small game development teams frequently struggle to determine which features justify their production cost, often relying on intuition rather than structured evaluation. This paper adapts the concept of Return on Time Invested (ROTI), originally used in agile meeting evaluation, into a method for complementing the assessment of feature production's efficiency in game development. The study is situated within a Research through Design process in which a team of three develops a comedic 3D driving game containing obstacles designed to amuse rather than alienate the player. Across two development iterations, semi-structured interviews with six playtesters per iteration are thematically coded into a normalized experiential value per obstacle and cross-referenced with development hours tracked in Jira. The findings reveal that experiential value and development cost are poorly correlated: the most expensive obstacle produced comparable player response to one costing 40% less development time, while the cheapest obstacle remained net negative across both iterations despite meaningful design improvement. Comedic framing that enhanced visual and auditory clarity functioned as an efficiency multiplier, raising experiential value while suppressing frustration. The paper concludes that normalized ROTI provides a lightweight, data-informed tool for identifying features that yield diminishing returns, offering small teams a shared framework for scoping conversations grounded in production data rather than assumption.

Keywords: comedy, game production, Lean project management, playtesting, Return on Time Invested, scope management.

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1 Introduction

Comedy-driven games rely on obstacles that amuse, but small development teams rarely have a systematic way of deciding which features justify their production cost. A mechanic that takes twenty hours to build but barely registers with players is a liability, especially in early development when scope decisions determine a project's success. This paper proposes a team-focused evaluation method for identifying such liabilities before they accumulate.

The study asks: how can Return on Time Invested (Derby & Larsen, 2006) be used to identify obstacles that yield diminishing returns of player satisfaction (*experiential returns*)? Return on Time Invested (ROTI) originates in agile project management as a method for evaluating whether time spent on an activity produced proportional value. This paper adapts the concept to game feature evaluation, cross-referencing coded playtest responses with tracked development hours to produce an efficiency score per obstacle.

The theoretical framework draws on Lean thinking (Ries, 2011), specifically the principles of waste elimination and the build-measure-learn cycle, which emphasizes rapid iteration, validated learning and the removal of effort that does not contribute enough . These principles are applied within a Research through Design (Coulton & Hook, 2017) process shared with two collaborators (see kappa, Section 1.2.1).

Data is collected across two iterations of a comedic 3D driving game. Each iteration is followed by semi-structured interviews with five to six playtesters, whose responses are thematically coded into an experiential value and cross-referenced with time-tracking from Jira. This paper does not evaluate whether the obstacles' mechanics or visuals are effective – those questions are addressed by the companion papers. The scope is strictly limited to production efficiency and feature scoping. The paper proceeds with Background, Research Design, Results and Discussion and Conclusion.

2 Background

2.1 Scope Management and Lean Thinking in Indie Development

Scope management is one of the most persistent challenges in indie game development. Small teams operate under tight time constraints and rarely have the resources to recover from overbuilt features. Francis (2018) illustrates this problem through a postmortem of his indie game *Heat Signature* (Suspicious Developments, 2017), which took three and a half years to develop – three times longer than his previous title, *Gunpoint* (Suspicious Developments, 2013). He reflects that much of that time was spent on features that contributed little to the final player experience, while the game's most well-received mechanics were among the cheapest to implement. This pattern, where production cost and experiential value are misaligned, is common in small-scale development but rarely quantified. Teams often rely on intuition or internal debate to decide what to cut, rather than on structured evaluation. The absence of a lightweight, data-informed tool for comparing feature cost against player response creates a gap that this paper aims to address.

Lean thinking offers a conceptual foundation for closing that gap. Ries (2011) outlines a framework for product development built on the elimination of waste and the rapid validation of assumptions through iterative build-measure-learn cycles. In this model, every feature is treated as a hypothesis: it is built in its simplest viable form, measured against real user responses and either refined or discarded based on what is learned. The framework was developed for startups operating under uncertainty, but the logic transfers directly to game development, where a demo or vertical slice functions as a minimum viable product and playtesting serves as the measurement step. Lean provides a mindset (prioritize learning, cut what does not deliver value), but it does not prescribe a specific metric for evaluating individual features against the time they consumed. It tells teams to eliminate waste but leaves them to judge *what waste looks like* on a per-feature basis.

2.2 Adapting ROTI to Feature Evaluation

Return on Time Invested (ROTI) is a metric that can help *identify* waste. Introduced by Derby and Larsen (2006) as a closing activity for agile retrospectives, ROTI asks participants to rate whether the time they spent in a meeting yielded proportional value, using a simple numerical scale. The method is subjective by design: it captures perceived value rather than financial return, and it is built for small-group evaluation with quick turnaround. These properties make it well-suited for adaptation to game production. In this paper, the ROTI concept is transferred from meeting evaluation to feature evaluation. Instead of asking whether a meeting was worth attending, the adapted method asks whether a game feature was worth building: meeting duration is replaced with tracked development hours, and participant ratings are replaced with thematically coded playtest responses. No prior work has applied ROTI in this way within game development, which positions this adaptation as the paper's primary contribution to the field.

3 Research Design

This study uses a mixed-methods approach embedded within the team's iterative Research through Design (RtD) process (Coulton & Hook, 2017). While the team shares a unified playtest structure and dataset, this paper analyzes the data strictly through a production management lens, cross-referencing development time costs with player responses to calculate the experiential ROTI (Derby & Larsen, 2006) per comedic obstacle.

3.1 Data Collection

Data collection draws on two sources: time-tracking data extracted from Jira, where all team members log development hours against obstacle-specific tasks, and semi-structured interviews conducted with playtesters after each iteration (see kappa, Section 4.3). The interview questions relevant to this paper targeted each obstacle's experiential impact: what the tester felt, whether the experience leaned toward amusement or frustration, and whether they would want to encounter the obstacle again (Appendix B).

3.2 Thematic Coding and ROTI Calculation

Interview responses are analyzed using thematic coding (Braun & Clarke, 2006). Each tester's response to each obstacle is assigned a code on a scale from –2 to +2. The code reflects the dominant sentiment expressed by the tester (Appendix B, Figure B1), and is accompanied by a short quote or paraphrase as justification. Codes for all testers are summed to produce an experiential value per obstacle, ranging from −12 to +12.

To enable comparison across obstacles with different tester counts and varying development scales, both experiential value and development time are normalized before calculating ROTI. Normalized Experiential Value (NEV) is calculated by dividing the summed experiential value by the maximum possible value for that group (±12 for six testers, ±10 for five), producing numbers from −1 to +1 where −1 represents unanimous negative response and +1 represents unanimous positive response. Normalized Time (NT) is calculated by dividing the obstacle's development hours by the average development hours across all obstacles and iterations (**20.34 hours**), where a value of 1.0 represents average effort. The normalized ROTI formula is **ROTI = NEV / NT**, where:

- NEV = experiential value / maximum possible experiential value;
- NT = development hours / average development hours across all entries.

A positive ROTI indicates that the obstacle delivered comedic value relative to its cost. A ROTI above 1.0 indicates exceptional efficiency. A near-zero ROTI indicates the investment producing negligible return. A negative ROTI indicates that the development time invested produced a net negative player experience. However, the formula is intended as a lightweight decision-support tool for structuring team scoping conversations, not as a precise measurement of feature value. The scores surface relative differences between obstacles rather than producing absolute values that can be referenced across projects.

4 Results

4.1 Iteration 1

By Week 2 of the production, two comedic obstacles were implemented: the banana peel and the trash bag (see Paper II, Appendix B). Six playtesters were interviewed using the per-obstacle semi-structured format described in the Research Design.

4.1.1 Player Response to Iteration 1

The banana peel produced a strongly negative experiential response. Four testers expressed clear frustration, describing the obstacle as "horrible," "terrible" and "too punishing" (Appendix C, Figure C1). Two testers who restarted the game multiple times attributed their restarts primarily to the banana peel. One tester treated it as a problem-solving challenge rather than a source of frustration, and one acknowledged the concept as fitting for the game's tone but felt the punishment was disproportionate. No tester reported finding the banana peel funny during gameplay, though two acknowledged the concept as inherently comedic in the abstract. The coded experiential value: −7.

The trash bag obstacle produced a near-neutral response (Appendix C, Figure C2). Two testers reacted positively, describing the 2D art as "cute" and the sound design as appealing. Two were mildly annoyed by the vision obstruction but noted it did not significantly impede gameplay since traffic intensity was low enough to compensate. Two expressed no strong reaction in either direction, with one describing themselves as "very neutral." No tester identified the trash bag as a major source of frustration. The coded experiential value: +1.

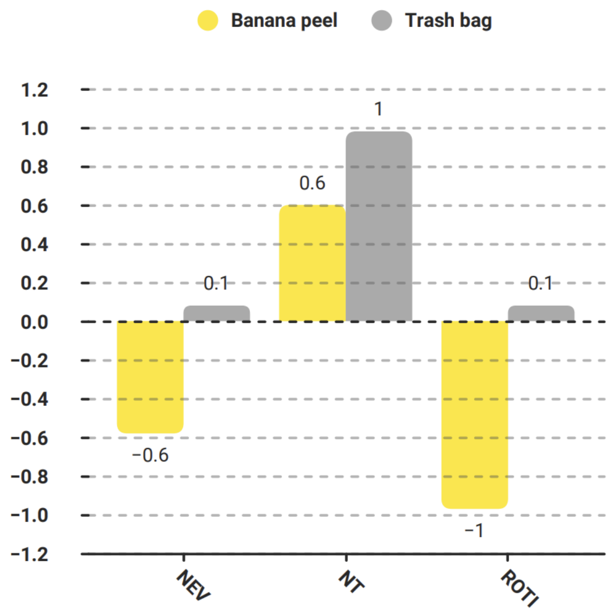
4.1.2 ROTI Calculation in Iteration 1

Jira time logs recorded 12.3 hours for the banana peel obstacle and 19.9 hours for the trash bag obstacle (Appendix A, Figures A1 and A2). Applying the formula to Iteration 1:

- Banana peel: NEV −0.58, NT 0.60, ROTI −0.97;
- Trash bag: NEV +0.08, NT 0.98, ROTI +0.08 (Figure 1).

Figure 1

Iteration 1 Impediments Comparison



The banana peel produced a negative ROTI, meaning the development time invested actively worsened the player experience. The trash bag produced a near-zero ROTI, indicating that nearly twenty hours of work resulted in a barely positive experiential return. Both obstacles failed to reach a positive ROTI that would justify their production cost, but they failed for different reasons. The banana peel's negative score stemmed from a design problem (the mechanic's punishment was disproportionate) rather than from excessive development time. The trash bag's near-zero score is a production efficiency issue: the experiential payoff was inoffensive but too mild to justify the time investment.

These findings informed the team's iteration priorities: the banana peel was flagged for design revision to reduce frustration without significant additional development time, while the trash bag was flagged for evaluation of whether further time investment could meaningfully raise its experiential value. Both adjustments are tested in Iteration 2.

4.2 Iteration 2

By Week 5 of the production, the team iterated on both existing obstacles and introduced the third, the octopus (see Paper II, Section 4.2). For this study, one additional playtester was recruited to get more answers on updated obstacles. Therefore, *six* playtesters reviewed the banana peel and the trash bag, and *five* playtesters reviewed the octopus.

4.2.1 Player Response to Iteration 2

The banana peel produced a milder response than in Iteration 1 (Appendix C, Figure C3). Three testers described it as mildly annoying but manageable, noting they could recover quickly or simply drive through without significant consequence. One tester found it slightly amusing, referring to how the car was affected by a banana peel. Two testers reported no strong reaction in either direction, describing the obstacle as functional but unremarkable. No tester reported the level of frustration seen in Iteration 1. The coded experiential value was −2, compared to −7 in Iteration 1.

The trash bag produced the strongest positive response of any obstacle across both iterations (Appendix C, Figure C4). Three testers reacted with clear amusement, citing the sound effect, the 2D sprite art and the absurdity of a fish or rat appearing on the windshield. Two testers described a positive but less emphatic reaction, finding it surprising and entertaining. One tester balanced amusement against gameplay impact, noting the trash bag was the most detrimental obstacle but that the comedic framing prevented genuine frustration. Multiple testers explicitly stated that the comedic presentation reduced how punishing the obstacle felt. The coded experiential value was +8.

The octopus produced a strongly positive experiential response among the five testers who encountered it (Appendix C, Figure C5). Four testers rated it positive comedic, describing it as "unexpected," "super goofy," and "very cute." One tester described the moment the octopus entered the car as genuinely funny, and another called it "the most unique obstacle." One tester coded as mildly negative, finding it the most frustrating obstacle due to its low visibility and the confusion it caused on every encounter. The coded experiential value was +7.

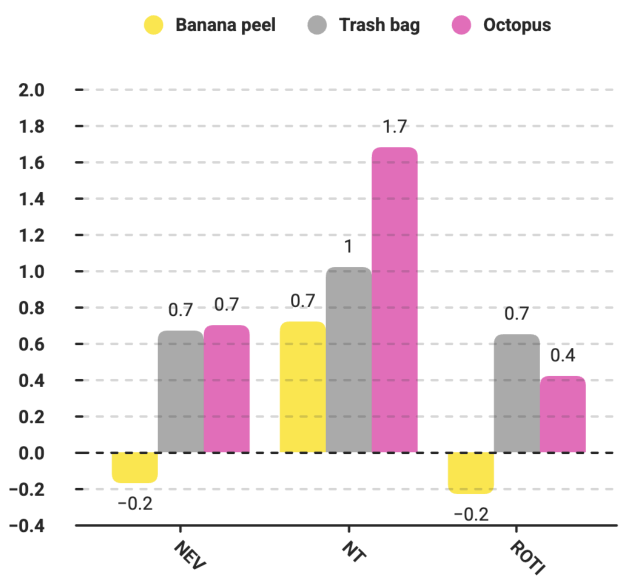
4.2.2 ROTI Calculation in Iteration 2

Jira time logs recorded 14.65 hours for the banana peel, 20.7 hours for the trash bag and 34.15 hours for the octopus (Appendix A, Figures A3–A5). Applying the normalized ROTI formula yields the following:

- Banana peel: NEV −0.17, NT 0.72, ROTI −0.23;
- Trash bag: NEV +0.67, NT 1.02, ROTI +0.65;
- Octopus: NEV +0.70, NT 1.68, ROTI +0.42 (Figure 2).

Figure 2

Iteration 2 Impediments Comparison



The trash bag achieved the highest ROTI of any obstacle across both iterations. Its experiential value was nearly identical to the octopus (+0.67 vs +0.70), but it required 40% less development time, producing a substantially higher efficiency score. The octopus, despite being the most positively received obstacle in raw experiential terms, was also the most expensive at 34.15 hours (over 1.6 times the dataset average) which diluted its ROTI. The banana peel improved from a ROTI of −0.97 in Iteration 1 to −0.23 in Iteration 2, indicating that the design revision meaningfully reduced frustration without significant additional development time (2.35 hours added between iterations). However, it remained the only obstacle with a negative ROTI, suggesting that further iteration or a fundamentally different approach would be needed to bring it into positive territory.

5 Discussion

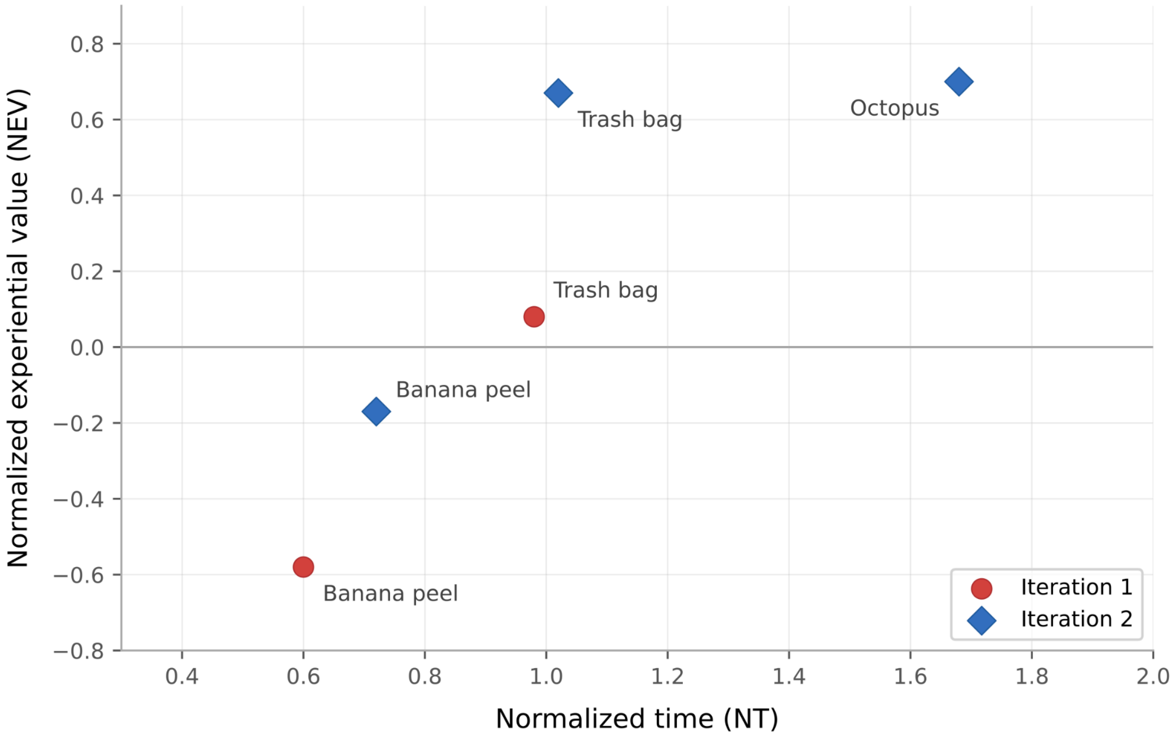
The research question asked how ROTI (Derby & Larsen, 2006) can be used to identify comedic obstacles that yield diminishing returns. The results suggest that the method does surface meaningful differences in production efficiency, though not always in the ways anticipated.

5.1 Cost–Value Misalignment

The most striking finding is that *experiential value and development cost are poorly correlated* (Figure 3). The octopus and trash bag produced nearly identical normalized experiential values (+0.70 and +0.67), yet the octopus required over 34 hours of development compared to the trash bag's 20.7 hours. The ROTI gap between them (+0.42 against +0.65) is a production cost outcome, not a design matter. This aligns directly with the pattern Francis (2018) describes in his *Heat Signature* postmortem: the best-received features were often among the cheapest to build, while the most time-consuming features did not proportionally improve the experience. The ROTI method makes this pattern visible at an early stage rather than in retrospective.

Figure 3

Normalized Experiential Value against Normalized Development Time across Both Iterations.



5.2 Improvement Without Proportional Cost

The banana peel's trajectory across iterations offers a different insight. Its ROTI improved from −0.97 to −0.23 with only 2.35 additional hours, most of which went toward adjusting the severity of the physics response. This suggests that low-ROTI obstacles are not necessarily candidates for removal – *a design revision can shift experiential value significantly without proportional time investment*. In Lean terms (Ries, 2011), the banana peel's Iteration 1 score represented validated learning: the build-measure-learn cycle confirmed that the obstacle's punishment was disproportionate to its comedic payoff, and the team acted on this signal. The method did not prescribe the fix, but it quantified the problem and later confirmed the improvement.

5.3 Comedy as Efficiency Multiplier

An unexpected finding was that *the comedic framing itself functioned as a production efficiency multiplier*. Multiple testers across both iterations stated that the trash bag's comedic presentation (the sound, the 2D sprite) actively reduced their perception of punishment. The trash bag's development cost included comparatively significant art and sound work, but this investment paid back: it raised the experiential value while suppressing frustration. The octopus, by contrast, was found funny, but its comedy did not offset the confusion caused by low visibility, which several testers flagged. Clarifying comedy is efficient, while confusing comedy is not.

6 Conclusion

These findings suggest that designers working under time constraints should prioritize obstacles with low implementation cost and strong sensory feedback (sound, visuals) over mechanically complex obstacles whose comedy depends on behaviour. The trash bag's high ROTI was driven not by sophisticated logic but by a distinctive sound effect and absurd 2D art – assets that were comparatively cheap to produce but disproportionately effective. On the opposite, the octopus required extensive scripting for its behavior and contextual triggering, yet produced a nearly identical experiential value. For teams deciding where to allocate their next sprint, ROTI data would favor producing two "trash bag-weight" features over one "octopus-weight" feature.

However, several limitations constrain these findings. Tester responses are self-reported and may not accurately reflect their in-game emotional experience. The sample of maximum six testers per iteration is small and affects the coding sensitivity. Said coding was performed by a single researcher, introducing coder bias. Additionally, as the project manager of the team, the author may carry a bias toward overestimating feature complexity, which could affect how production data is interpreted. The reliability of time-tracking data depends on teammates logging hours consistently and honestly. The ROTI formula weights all development hours equally, though some hours may contribute more to experience than others. The normalized time denominator is derived from this dataset alone and would shift with additional data points. Finally, only three obstacles were evaluated, limiting the comparison. These limitations are acknowledged as inherent to a small-scale, practice-based undergraduate study working under real production constraints.

This paper contributes a lightweight complementary method for evaluating feature efficiency. It does not replace design intuition or playtester feedback but offers a structured addition to both. For small teams operating under time pressure, ROTI provides a shared language for structuring conversations, replacing subjective feature potential judging with efficiency assessment. Future work could refine the method by separating programming, art and sound hours to identify which type of work contributes more to experiential return, and by applying ROTI across a larger obstacle set or in a different genre to test its transferability.

7 References

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3(2)*, 77–101. https://doi.org/10.1191/1478088706qp063oa

Coulton, P., & Hook, A. (2017). Games design research through game design practice. In P. Lankoski & J. Holopainen (Eds.), *Game design research: An introduction to theory & practice* (pp. 97–116). ETC Press.

Derby, E., & Larsen, D. (2006). *Agile Retrospectives: Making Good Teams Great*. Pragmatic Bookshelf.

Francis, T. (2018, March 19–23). *Dealing with scope change in "Heat Signature" and "Gunpoint"* [Conference presentation]. Game Developers Conference, San Francisco, CA, United States. https://gdcvault.com/play/1024932/Dealing-with-Scope-Change-in

Ries, E. (2011). *The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses*. Crown Business.

7.1 Ludography

Suspicious Developments. (2013). *Gunpoint* [Video game].

Suspicious Developments. (2017). *Heat Signature* [Video game].

Appendix A – Jira Time Tracking

This appendix presents development time data extracted from the team's Jira board. Hours are logged by team members against sub-tasks tied to each obstacle's User Story. Figures A1–A2 correspond to Iteration 1 and Figures A3–A5 to Iteration 2. Total hours per obstacle serve as the denominator in the normalized ROTI formula described in section 4.

Figure A1

Tracked Hours for Banana Peel, Jira (Iteration 1)

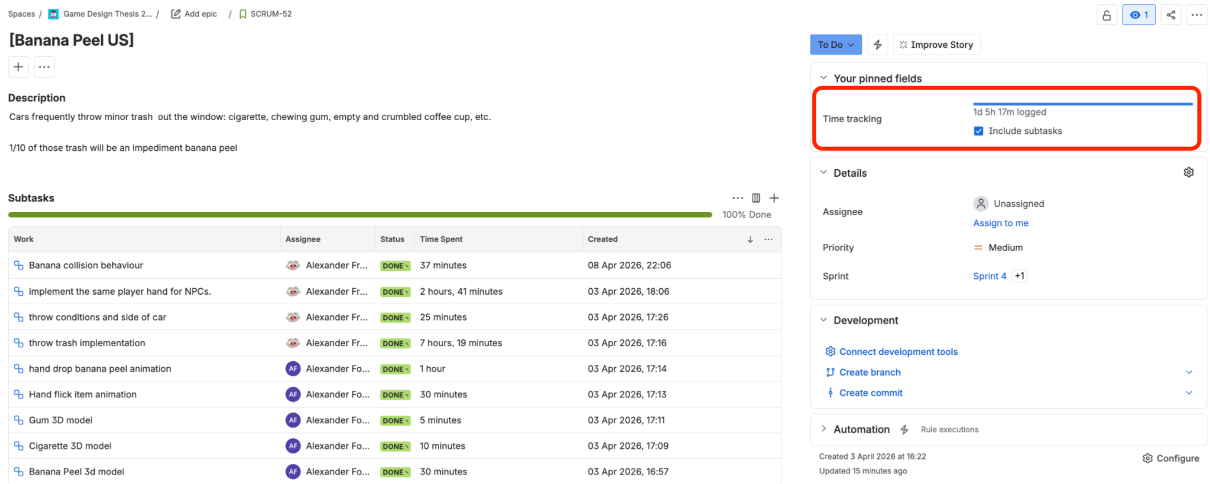


Figure A2

Tracked Hours for Trash Bag, Jira (Iteration 1)

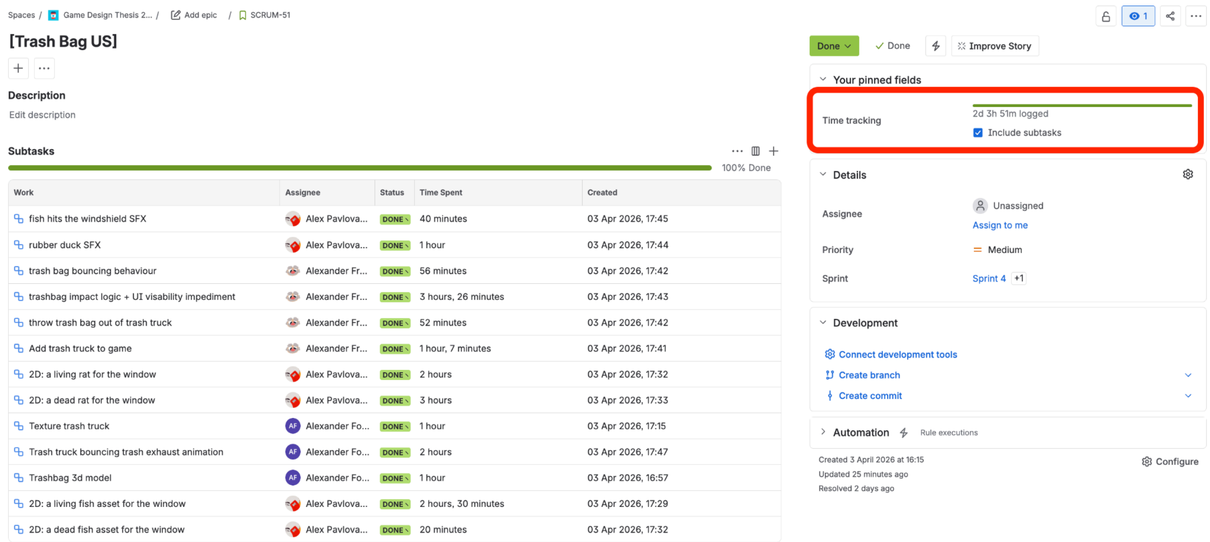


Figure A3

Tracked Hours for Banana Peel, Jira (Iteration 2)

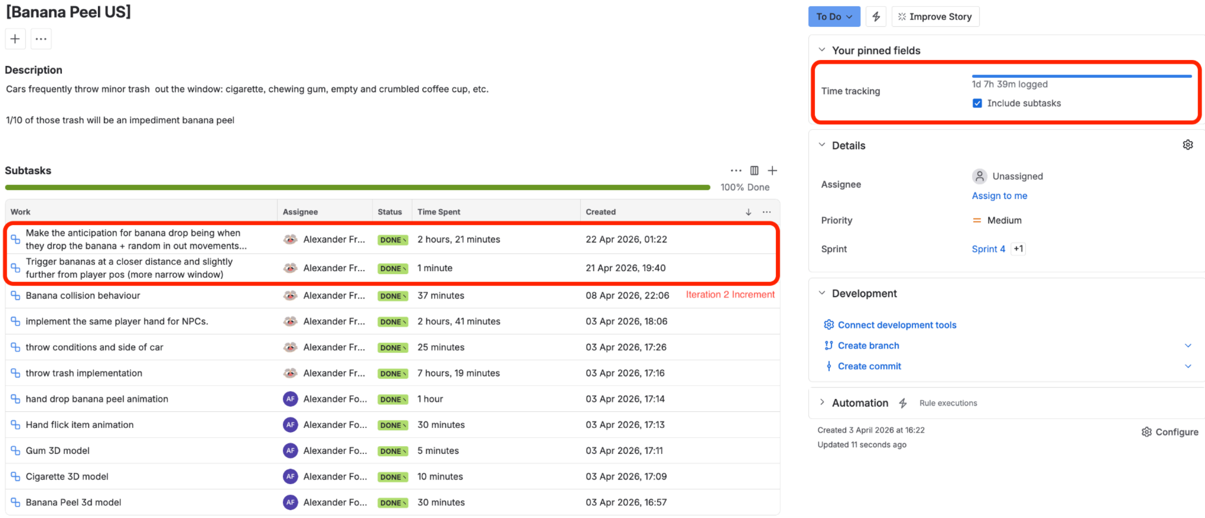


Figure A4

Tracked Hours for Trash Bag, Jira (Iteration 2)

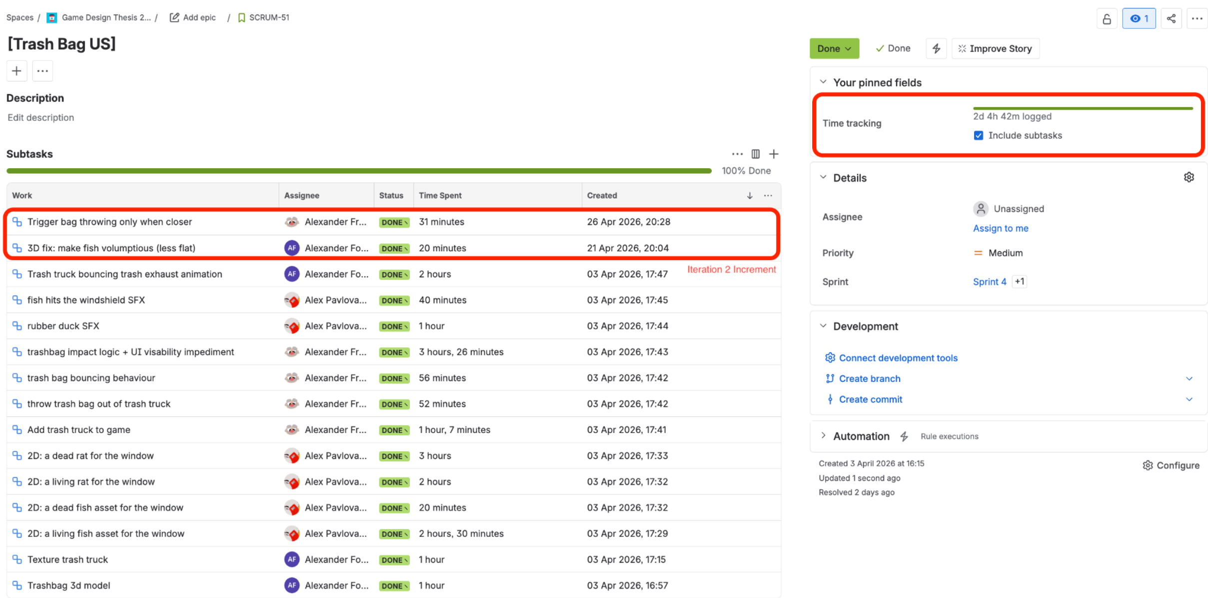
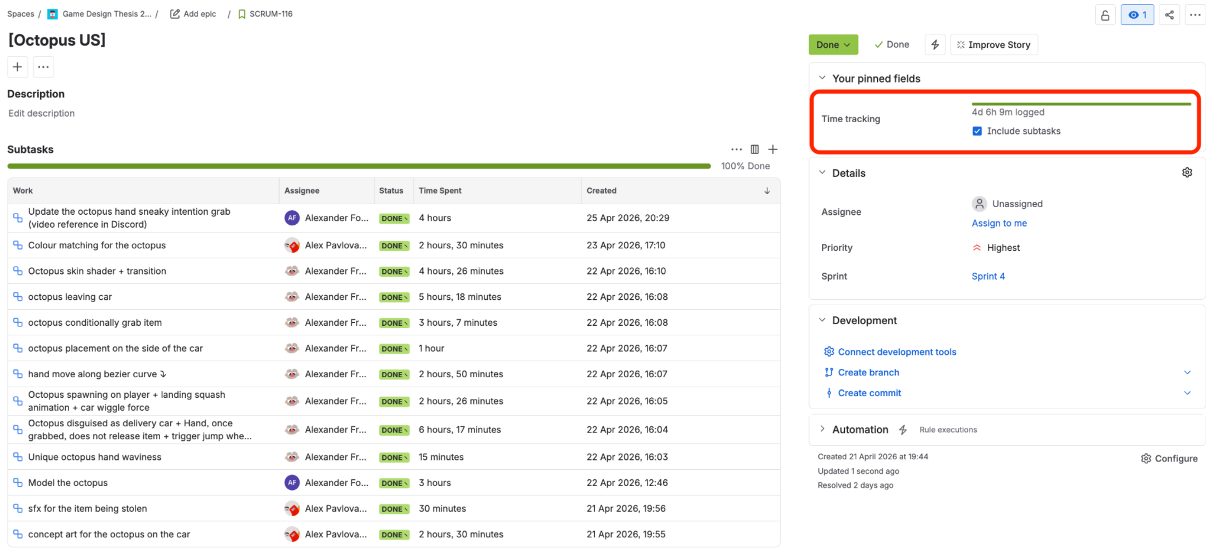


Figure A5

Tracked Hours for Octopus, Jira (Iteration 2)



Appendix B – Semi-Structured Interview Guide

The following questions were asked per obstacle during semi-structured interviews conducted after each playtest session. Testers were shown the screenshot of each obstacle before responding. Follow-up questions were used to clarify or deepen responses but did not introduce new topics. The interviewer did not lead testers toward specific answers.

Per-obstacle questions:

1. What did you feel when this obstacle happened to you?
2. Did it make you laugh, annoy you, or both?
3. Would you like to encounter this obstacle again in the next version of the game?
4. If you were to compare this obstacle to [other obstacle], do you have any comments on how they are similar or different in terms of how they felt?

Coding scale applied to responses:

- Each tester's response to each obstacle was assigned a single code reflecting the dominant sentiment expressed;
- Codes were assigned by a single researcher after reviewing interview recordings.

Figure B1

Coding Guide

Code	***Lable***	***Description***
+2	Positive comedic	Tester laughed, expressed delight, described the obstacle as funny, surprising, or memorable in a positive way
+1	Mild positive	Tester smiled, found it clever or acceptable, described it as fine or mildly amusing
0	Neutral	Tester had no strong reaction, did not notice the obstacle, or could not remember it
-1	Mild negative	Tester found it slightly annoying, confusing, or disruptive but not strongly negative
-2	Negative	Tester expressed clear frustration, called it unfair, said it broke the flow, or described it as not fun

Appendix C – Coding Breakdown

This appendix documents the thematic coding applied to semi-structured interview responses collected after each playtest. Each tester's dominant sentiment toward each obstacle is assigned a single code on a five-point scale from −2 (negative) to +2 (positive comedic), accompanied by a short quote or paraphrase as justification. The coding scale is defined in Appendix B. Summed codes per obstacle produce the experiential value used as the numerator in the normalized ROTI formula.

Figure C1

Coding Breakdown for Banana Peel, Iteration 1

Tester	***Code***	***Evidence***
T1	–1	Frustration from losing control; called it "too punishing" but wants it back.
T2	+1	Treated it as problem-solving; "I wasn't getting annoyed".
T3	–2	"I felt hella frustrated"; restarted four times.
T4	–1	Heavily impacted; high frustration noted.
T5	–2	"Horrible to drive into"; would not want to encounter again.
T6	–2	"Anger"; "...the [censored] banana peel ruined everything".
Total		**–7**

Figure C2

Coding Breakdown for Trash Bag, Iteration 1

Tester	***Code***	***Evidence***
T1	–1	"Annoyance and frustration" from blocked vision and dropped items; acknowledged the concept had "a bit more humor" than banana peel but frustration dominated.
T2	–1	"Definitely annoying"; valued the art and sound but felt it "takes away all your agency".
T3	0	"Very neutral"; effects passed too quickly to register emotionally; did not feel gameplay-impacted compared to banana peel.
T4	0	Moderate frustration but view obstruction "wasn't that much of an issue because of the traffic being low intensity, you can slow down with no

		losses".
T5	+2	"I thought it was cute art"; "cute and cool way of showcasing that you're half blind"; "more so funny"; "didn't feel too frustrating at all".
T6	+1	"It was nice"; liked the sound and visuals; wanted no changes; no frustration expressed.
Total		**+1**

Figure C3

Coding Breakdown for Banana Peel, Iteration 2

Tester	***Code***	***Evidence***
T7	–1	"Didn't really make me laugh... more annoying"; comedy had no mitigating effect; described it as "a normal obstacle, not like an unexpected joke".
T8	–1	"Felt a bit frustrated because I didn't have control of the car"; banana was "a bit more annoying" than other obstacles; would remove it if not tweaked.
T9	–1	"I did feel like it was a negative thing"; not strongly frustrated but overall impression negative; kept driving without major disruption.
T10	0	"It just was there"; no emotional reaction; "banana peel is actually in a good spot"; acknowledged the genre is not his preference.
T11	0	Both laughed and annoyed; "pretty easy... I just kind of was sent to the side and I just put the car back"; balanced, no dominant sentiment.
T12	+1	"I was fine with it"; "kind of funny that my [big] car is influenced by this little banana peel"; comedy slightly reduced punishment perception.
Total		**–2**

Figure C4

Coding Breakdown for Trash Bag, Iteration 2

Tester	***Code***	***Evidence***
T7	+2	"It was funny, the sound effect and silly sprites"; laughed; named it most comedic obstacle; "silliest of them all, probably decreased frustration".
T8	+1	"Surprising" and "nice with the sound"; 2D felt slightly out of place but positively surprising; no frustration reported.
T9	0	Called it "most detrimental to gameplay" but "some kind of large rodent, I’m not going to get upset about that"; comedy offset impact.
T10	+2	"That was funny, unexpected"; "most laugh"; zero frustration; didn’t even try to avoid it.
T11	+1	Both laughed and got annoyed; the sound effect was funny; comedy and visuals "made it less frustrating"; they wanted to encounter it again.
T12	+2	"Wasn’t annoyed. It was funny"; could "appreciate the rat on [their] window"; comedy reduced punishment; wanted "more trash bags".
Total		**+8**

Figure C5

Coding Breakdown for Octopus, Iteration 2

Tester	***Code***	***Evidence***
T7	–1	"Most frustrating one because hardest to notice"; confused on every encounter; comedy did not reduce punishment.
T8	+2	Most memorable and most funny obstacle; smiled when it entered the car; "it was just funny to me".
T9	+2	Octopus made them smile first; "I got a friend now"; positive experience throughout.
T11	+2	"So unexpected... design is amazing, super goofy, I love it"; "more funny than punishing"; comedy explicitly reduced punishment.
T12	+2	"Very cute and very funny"; "loved that it stole my flower"; surprised in a good way; called it "just a thief" and laughed.
Total		**+7**